Notes: Le Grand and Ragot (JPE, 2025)

Optimal Fiscal Policy with Heterogeneous Agents and Capital: Should We Increase or Decrease Public Debt and Capital Taxes?

Contents

1 Big picture

- **Question.** In a heterogeneous-agent incomplete-markets economy with capital accumulation, what are the optimal steady-state levels of capital taxes and public debt, and how should taxes and debt respond after a public spending shock?
- **Setting.** The government finances exogenous public spending with three instruments: public debt, a tax on capital income, and a progressive labor-income tax. Households are heterogeneous, face idiosyncratic labor-income risk, and may be credit constrained.
- **Core challenge.** The classical Chamley–Judd logic points toward a zero long-run capital tax. But in incomplete-markets environments, agents save for self-insurance, credit constraints may bind, and the planner must simultaneously manage redistribution, insurance, and the tax base.
- **Main theoretical result.** The steady-state optimal capital tax can be **positive** when credit constraints occasionally bind. However, this is not automatic: existence and positivity depend on the shape of preferences. The key force is a **price externality of savings** on posttax factor prices.
- **Main debt-dynamics result.** After a public spending shock, the optimal response of public debt depends crucially on persistence. When the shock is **transitory**, debt rises to help smooth consumption; when the shock is **persistent**, debt falls (or rises much less), because future tax distortions needed to stabilize debt become too costly.
- **Main quantitative result.** In a calibrated HANK model disciplined by the actual US fiscal system through an inverse-optimal approach, the qualitative theory survives: capital taxes and labor-tax progressivity rise after a spending shock, while debt rises for low persistence and falls for high persistence.
- **Why this paper matters.** It gives a tractable link between optimal-tax theory, incomplete markets, and practical fiscal dynamics. The paper is especially useful for thinking about debt finance, redistribution, and fiscal stabilization when households self-insure and credit constraints bind.

2 Data and measurement (key objects)

This is primarily a theory-and-quantitative paper rather than a reduced-form empirical paper, so the relevant "data" objects are the model primitives, calibration targets, and state variables.

2.1 Observed state variables and aggregate objects

- **Time.** Discrete time, indexed by $t = 0, 1, 2, \ldots$.
- **Agents.** A continuum of heterogeneous agents, indexed by i, partitioned into ex ante types $f = 1, \ldots, F$, with population shares m_f.
- **Assets.** Private savings $a^f_{i,t}$, aggregate private wealth A_t, physical capital K_t, and public debt B_t.
- **Production.** Aggregate output Y_t produced by a representative firm using capital and labor.

- **Prices and fiscal tools.** Pretax wage $\tilde{w}_t$, pretax interest rate $\tilde{r}_t$, posttax wage w_t, gross posttax return R_t, capital tax t_t^K, and labor-tax parameters (k_t, τ_t).
- **Public spending.** An exogenous aggregate path $(G_t)_{t\geq 0}$.

2.2 Idiosyncratic risk, heterogeneity, and timing

- **Idiosyncratic productivity.** Agents differ through labor-productivity processes $y_{i,t}^f$ that follow finite-state Markov chains.
- **Credit constraint.** Agents face
$$a_{i,t}^f \geq -\underline{a},$$
with the simple analytical model later setting $\underline{a} = 0$.
- **MIT shock timing.** The whole future path of the aggregate shock is revealed at date 0 and the planner cannot renege on past commitments. Therefore the paper studies transition dynamics after an MIT shock under full commitment.
- **Unobservable object in welfare analysis.** The planner attaches social welfare weights q_f to different ex ante types. In the quantitative model, these are not imposed arbitrarily but recovered with an inverse-optimal approach.

2.3 Simple analytical economy

The tractable analytical model adds the following restrictions:

- one productivity process only ($F = 1$),
- two productivity states, employed and unemployed,
- an antidiagonal transition matrix, so agents alternate deterministically between employment and unemployment,
- a linear labor tax ($\tau_t = 0$),
- zero borrowing limit ($\underline{a} = 0$).

This simple environment is crucial because it isolates the role of binding credit constraints and the price externality of savings without the full computational burden of a richer HANK model.

2.4 Quantitative calibration targets

The quantitative model is disciplined by US moments and policy targets.

- **Technology/household parameters.** The period is a quarter. The discount factor and technology are chosen to match an annual capital-output ratio of 2.7. The capital share is $\alpha = 0.36$, depreciation is $\delta = 2.5\%$, the Frisch elasticity is $J = 0.5$, and the labor-supply scaling parameter is $\chi = 0.05$.
- **Ex ante types.** There are three types, motivated by education groups in the US labor force, with average productivities 0.8, 1.0, and 2.0, and population shares of one-third each.

- **Idiosyncratic risk processes.** The three type-specific AR(1) productivity processes have persistence $(0.986, 0.98, 0.98)$ and standard deviations $(0.16, 0.132, 0.132)$ before discretization.
- **Fiscal targets.** The benchmark steady-state fiscal system is chosen to match US values: capital tax 36%, labor-tax progressivity $\tau = 0.18$, labor-tax scaling $k = 0.75$, debt-to-output ratio $B/Y = 61\%$, and public-consumption share $G/Y = 17\%$.

3 Models and empirical specifications (all equations + interpretation)

3.1 Production, assets, and fiscal instruments

Output is produced with a Cobb–Douglas technology,

$$Y_t = F(K_{t-1}, L_t) = K_{t-1}^{\alpha} L_t^{1-\alpha} - \delta K_{t-1}.$$

The representative firm rents capital and labor competitively, so pretax factor prices satisfy

$$\tilde{w}_t = F_L(K_{t-1}, L_t), \qquad \tilde{r}_t = F_K(K_{t-1}, L_t).$$

Labor income is taxed through the Heathcote–Storesletten–Violante schedule

$$T_t(\tilde{w}yl) = \tilde{w}yl - k_t(\tilde{w}yl)^{1-\tau_t},$$

so posttax labor income equals

$$\tilde{w}_t y_{i,t}^f l_{i,t}^f - T_t(\tilde{w}_t y_{i,t}^f l_{i,t}^f) = w_t (y_{i,t}^f l_{i,t}^f)^{1-\tau_t},$$

where

$$w_t \equiv k_t \tilde{w}_t^{1-\tau_t}.$$

The gross posttax return on savings is

$$R_t = 1 + r_t = 1 + (1 - t_t^K)\tilde{r}_t.$$

Interpretation. The planner does not directly choose allocations; it chooses a tax system and debt path that decentralize an allocation. The capital tax distorts intertemporal saving, while the labor-tax level and progressivity distort labor supply and the income distribution.

3.2 Household problem in the general model

Each (i, f) agent solves

$$\max_{\{c_{i,t}^f, l_{i,t}^f, a_{i,t}^f\}_{t\geq 0}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t U(c_{i,t}^f, l_{i,t}^f)$$

subject to

$$c_{i,t}^f + a_{i,t}^f = R_t a_{i,t-1}^f + w_t (y_{i,t}^f l_{i,t}^f)^{1-\tau_t},$$

$$a_{i,t}^f \geq -\underline{a}, \qquad c_{i,t}^f \geq 0, \qquad l_{i,t}^f \geq 0.$$

The Euler equation is

$$U_c(c_{i,t}^f, l_{i,t}^f) = \beta \mathbb{E}_t\Big[R_{t+1} U_c(c_{i,t+1}^f, l_{i,t+1}^f)\Big] + n_{i,t}^f,$$

where $n_{i,t}^f \geq 0$ is the multiplier on the borrowing constraint. The labor first-order condition is

$$-U_l(c_{i,t}^f, l_{i,t}^f) = (1-\tau_t) w_t y_{i,t}^f (y_{i,t}^f l_{i,t}^f)^{-\tau_t} U_c(c_{i,t}^f, l_{i,t}^f).$$

Market clearing requires

$$A_t = K_t + B_t, \qquad \sum_{f=1}^{F} m_f \int y_{i,t}^f l_{i,t}^f \, \mathrm{d}\ell_f(i) = L_t,$$

and goods-market feasibility is

$$\sum_{f=1}^{F} m_f \int c_{i,t}^f \, \mathrm{d}\ell_f(i) + G_t + K_t = K_{t-1} + F(K_{t-1}, L_t).$$

Interpretation. In incomplete markets, private savings serve both productive investment and self-insurance. This is what makes public debt potentially useful even when it crowds into the same one-period safe asset that households want to hold.

3.3 Planner objective and Ramsey equilibrium

The planner maximizes a weighted sum of lifetime utilities,

$$W_0 = \sum_{f=1}^{F} m_f q_f \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t \int_{i \in I_f} U(c_{i,t}^f, l_{i,t}^f) \, \mathrm{d}\ell_f(i) \right],$$

with weights normalized so that

$$\sum_{f=1}^{F} m_f q_f = 1.$$

A Ramsey equilibrium is the competitive equilibrium generated by the fiscal policy $(t_t^K, k_t, \tau_t, B_t)_{t \geq 0}$ that maximizes W_0.

Interpretation. The planner chooses taxes and debt, not firm or household actions directly. The social weights summarize redistributive preferences over ex ante types.

3.4 The tractable two-state model

To derive analytical results, the paper studies a simplified setting with one ex ante type and two productivity states: employed (e) and unemployed (u). The planner's program can then be written as

$$\max_{\{c_{e,t}, c_{u,t}, a_{e,t}, a_{u,t}, l_{e,t}, B_t, A_t, R_t, w_t\}_{t \geq 0}} \sum_{t=0}^{\infty} \beta^t \big(U(c_{e,t}, l_{e,t}) + U(c_{u,t}, 0) \big)$$

subject to

$$c_{e,t} + a_{e,t} = R_t a_{u,t-1} + w_t l_{e,t},$$

$$c_{u,t} + a_{u,t} = R_t a_{e,t-1},$$
$$U_c(c_{e,t}, l_{e,t}) = \beta R_{t+1} U_c(c_{u,t+1}, 0),$$
$$U_c(c_{u,t}, 0) \geq \beta R_{t+1} U_c(c_{e,t+1}, l_{e,t+1}),$$
$$-U_l(c_{e,t}, l_{e,t}) = w_t U_c(c_{e,t}, l_{e,t}),$$

plus the government budget constraint, asset-market clearing, and nonnegativity/borrowing constraints.

Interpretation. In this simple model, only the unemployed can be credit constrained in equilibrium. The employed save, the unemployed dissave or hit the borrowing constraint, and the planner trades off tax smoothing, insurance, and redistribution very transparently.

3.5 Main characterization of the stationary Ramsey equilibrium

In a stationary Ramsey equilibrium (SRE) with binding credit constraints for the unemployed, Proposition 1 establishes two central conditions:

$$1 + F_K = \frac{1}{\beta},$$

which is the **modified golden rule**, and

$$1 - \beta R = \frac{F_L - w}{w} \cdot \frac{\jmath_u - \jmath_e + \varsigma^l_{c,e}}{\jmath_e + 1/\mathcal{J}_e - \varsigma^l_{c,e} + \varsigma^c_{l,e}}.$$

Here:

- $\jmath_e$ and $\jmath_u$ are inverses of the intertemporal elasticities of substitution for employed and unemployed agents,
- $\mathcal{J}_e$ is the Frisch elasticity of labor supply,
- $\varsigma^l_{c,e}$ and $\varsigma^c_{l,e}$ capture nonseparability between consumption and labor.

The paper then rewrites the planner's FOCs as

$$1 - \beta R = \Xi\big(\jmath_u - \jmath_e + \varsigma^l_{c,e}\big),$$
$$\frac{F_L - w}{w} = \Xi\Big(\jmath_e + \frac{1}{\mathcal{J}_e} - \varsigma^l_{c,e} + \varsigma^c_{l,e}\Big),$$

for some positive multiplier Ξ.

Interpretation. This is the heart of the paper. The planner sets aggregate savings and labor supply while internalizing how those choices move factor prices. More private saving expands the capital-tax base, but it also moves posttax returns and wages in a way that matters for redistribution and self-insurance. A positive capital tax arises when the planner wants to correct this savings externality.

3.6 Why CRRA gives zero capital tax, but GHH or DRRA can give positive capital tax

For separable CRRA utility, the inverse IES is constant across employed and unemployed agents, so the savings externality disappears. Corollary 1 states that in this case the steady-state capital tax is zero.

For GHH preferences,

$$U(c,l) = u\left(c - \chi^{-1}\frac{l^{1+1/J}}{1+1/J}\right),$$

which kills the wealth effect on labor supply. The paper shows that the wedges satisfy

$$1 - \beta R = \frac{F_L - w}{w} J\,\jmath\,(1+\beta-\beta R)^{1/\jmath - 1}.$$

In the log-GHH case ($\jmath = 1$), this simplifies to a tight relation between the capital and labor tax:

$$(1-\beta)t^K = \frac{t^L}{1-t^L} J(1+\beta).$$

Interpretation. Under GHH, once labor taxation is positive, capital taxation is positive too. So the model overturns Chamley–Judd not because markets are incomplete per se, but because incomplete markets interact with the right utility shape to make the savings externality quantitatively relevant.

3.7 Existence of the steady-state Ramsey equilibrium

The paper first defines a threshold γ_1 such that if

$$G \leq \gamma_1 Y^{FB},$$

the first-best steady state can be decentralized with zero taxes and perfect consumption smoothing.

When public spending is larger than that, the first best cannot be sustained because it would require the government to hold too much capital. Proposition 3 then introduces two additional thresholds, γ_{SW} and γ_{La}, and shows that when

$$\gamma_1 Y^{FB} < G \leq \min\{\gamma_{\mathrm{SW}}, \gamma_{\mathrm{La}}\} Y^{FB},$$

there exists a unique SRE with binding credit constraints and strictly positive labor and capital taxes.

Interpretation.

- γ_1 is the non-first-best threshold.
- γ_{SW} is the **Straub–Werning condition**: public spending cannot be so high that the planner prefers a nonstationary shrinking-capital path.
- γ_{La} is a **Laffer-type condition**: tax bases must remain large enough to finance the required spending.

The paper also establishes a threshold γ^{pos} such that public debt is positive if

$$\gamma_1 \leq 0 \quad \text{and} \quad G \leq \gamma^{\text{pos}} Y^{FB}.$$

Interpretation. Positive capital taxation and positive public debt can coexist because the planner wants households to keep saving for insurance, but does not want all that saving to become private capital. Public debt absorbs excess desired saving while the capital tax limits the price externality of savings on factor prices.

3.8 Dynamics after a public spending MIT shock

Public spending follows the MIT-shock process

$$\hat{G}_t = \begin{cases} \hat{G}_0, & t = 0, \\ \rho_G \hat{G}_{t-1}, & t > 0. \end{cases}$$

The optimal capital stock evolves as

$$\hat{K}_t = r_K \hat{K}_{t-1} + j_K \hat{G}_t, \qquad r_K > 0,\ j_K < 0,$$

so a positive spending shock lowers capital. Stability requires the Blanchard–Kahn condition

$$\alpha \leq \frac{1}{1 + (1-\beta)(1+J)}.$$

The paper derives closed-form impulse responses for capital and debt. In particular,

$$\hat{B}_t = \hat{G}_0 \big(\Theta_K r_K^t - \Theta_G \rho_G^t \big).$$

With debt positive in steady state, Proposition 5 shows

$$\frac{\partial \hat{B}_0}{\partial \rho_G}\bigg|_{\hat{G}_0} < 0,$$

and, when comparing shocks with the same net present value,

$$\frac{\partial \hat{B}_0}{\partial \rho_G}\bigg|_{\widehat{NPV}_0} < 0.$$

Interpretation. Persistence is the key state variable for debt finance. If the shock is temporary, more debt is useful because it smooths taxes and consumption. If the shock is persistent, issuing more debt today implies highly persistent future taxes to stabilize it, exactly when capital and output are already depressed. Then the planner front-loads adjustment through taxes instead.

3.9 Ex ante heterogeneity and social weights

The analytical model is then extended to two ex ante types. Type A faces unemployment risk and saves; type B is always employed and does not save. Let

$$\Lambda \equiv \frac{\Omega_B w y^B l_e^B}{\Omega_A w y^A l_e^A} = \frac{\Omega_B (y^B)^{J+1}}{\Omega_A (y^A)^{J+1}}.$$

Proposition 6 shows that the wedges satisfy

$$\frac{1-\beta R}{q_A} = \frac{q_B}{q_A} - (1+\beta)\Lambda + \frac{F_L - w}{w} J(1+\beta)(1+\Lambda),$$

or equivalently,

$$\frac{(1-\beta)t^K}{q_A} = \frac{q_B}{q_A} - (1+\beta)\Lambda + J(1+\beta)(1+\Lambda)\frac{t^L}{1-t^L}.$$

Interpretation. Social weights and optimal taxes are jointly determined. Raising the planner's weight on the non-saving, always-employed type makes labor taxation less attractive relative to capital taxation. This is the analytical foundation for the inverse-optimal exercise in the quantitative model.

3.10 The general model and transformed Ramsey program

With many ex ante types and finite-state Markov income processes, the paper uses a transformed planner problem. For GHH preferences, labor can be written as

$$l_{i,t}^f = l_t \, (y_{i,t}^f)^{(1-\tau_t)/(1/J+\tau_t)},$$

with the unitary labor supply

$$l_t \equiv [\chi(1-\tau_t)w_t]^{1/(1/J+\tau_t)}.$$

The paper then defines the transformed tax-progressivity object

$$\tilde{\tau}_t \equiv (1/J+1)\frac{1-\tau_t}{1/J+\tau_t},$$

and transformed utility argument

$$x_{i,t}^f \equiv c_{i,t}^f - \chi^{-1}\frac{(l_{i,t}^f)^{1+1/J}}{1+1/J}.$$

The Ramsey program becomes a sequential maximization over

$$\{r_t, \tilde{\tau}_t, B_t, K_t, L_t, l_t, (a_{i,t}^f, x_{i,t}^f, n_{i,t}^f)\}_{t\geq 0},$$

subject to a transformed government budget constraint, transformed household budget constraints, Euler equations, borrowing constraints, and market clearing.

Interpretation. The transformation exploits GHH preferences to reduce dimensionality and makes the planner's FOCs interpretable in terms of social values of liquidity and public funds.

3.11 Planner FOCs in the general model

The key new object is the **marginal social value of liquidity**

$$\omega_{i,t}^f \equiv q_f u'(x_{i,t}^f) - \big(\lambda_{i,t}^f - (1+r_t)\lambda_{i,t-1}^f\big)u''(x_{i,t}^f),$$

where $\lambda_{i,t}^f$ is the multiplier on the household Euler equation. The marginal value of public funds raised from agent (i,f) is

$$\widehat{\omega}_{i,t}^f \equiv m_t - \omega_{i,t}^f,$$

where m_t is the multiplier on the government budget.

The planner's public-funds smoothing condition is

$$\widehat{\omega}_{i,t}^f = \beta \mathbb{E}_t\big[(1+r_{t+1})\widehat{\omega}_{i,t+1}^f\big].$$

The FOC for public debt is

$$m_t = \beta(1+\tilde{r}_{t+1})m_{t+1},$$

which again implies the modified golden rule in steady state.

The capital-tax FOC equates a **net distributive gain** from taxing wealth to a **cost on saving incentives**:

$$\sum_f m_f \int \widehat{\omega}_{i,t}^f a_{i,t-1}^f \, \mathrm{d}\ell_f(i) = \sum_f m_f \int \lambda_{i,t-1}^f u'(x_{i,t}^f) \, \mathrm{d}\ell_f(i).$$

The labor-tax level and progressivity FOCs have analogous "benefit versus incentive cost" structure.

Interpretation. The general HANK model preserves the same logic as the simple model: fiscal instruments are used to trade off distributive gains, tax-base effects, and the way taxes change labor and saving incentives across heterogeneous households.

3.12 Quantitative implementation and inverse-optimal mapping

The quantitative strategy has three steps.

- **Step 1.** Calibrate technology, preferences, productivity risk, and the steady-state US fiscal system.
- **Step 2.** Use the planner's steady-state FOCs to infer the social welfare weights q_f for which that observed fiscal system is optimal.
- **Step 3.** Starting from that steady state, simulate optimal responses to MIT shocks of varying persistence.

Because fiscal policy has four instruments but the government budget constraint removes one degree of freedom and the public-debt FOC pins down the steady-state return without restricting social weights, only two FOC restrictions identify the welfare weights. With the normalization $\sum_f m_f q_f = 1$, the paper therefore sets $F = 3$ types so that the social weights are exactly identified.

The model is solved with the truncation method of Le Grand and Ragot, refined so that a finite set of 455 histories approximates the heterogeneous-agent distribution well.

4 Identification and instruments (what makes the estimates credible)

This is a theoretical and quantitative-identification paper, so "identification" means what pins down the model's economic mechanisms and what lets the quantitative exercise separate the relevant forces.

4.1 What fails in the standard approach

- **Chamley–Judd logic is not robust here.** In complete markets or in incomplete-markets models with separable CRRA utility, steady-state capital taxes are driven to zero.

- **A naive extension is wrong.** It is not enough to say "markets are incomplete, therefore capital taxes are positive." The paper shows that incomplete markets are necessary but not sufficient.
- **Why this matters.** Without extra structure, one might mistakenly interpret any positive tax result as a generic HANK implication. The paper shows that utility curvature, labor-supply interactions, and the existence of standard SREs are all crucial.

4.2 What identifies the positive-capital-tax result

- **Occasionally binding credit constraints** create a role for private saving as self-insurance.
- **Savings externality.** More saving changes posttax factor prices. The planner internalizes this price externality when setting taxes.
- **Utility shape.** The externality is zero in separable CRRA but generally nonzero in DRRA or GHH environments. This is what identifies when a positive capital tax can appear.

4.3 What identifies existence of the steady-state Ramsey equilibrium

The existence result is pinned down by four conditions.

- **Non-first-best condition.** Public spending must be high enough that the first-best allocation cannot be financed by the government simply holding enough capital.
- **Straub–Werning condition.** Public spending cannot be so high that the planner prefers a nonstationary shrinking-capital path.
- **Laffer condition.** The tax system must still raise enough revenue.
- **Blanchard–Kahn condition.** The transition system must be dynamically stable.

Interpretation. The paper is unusually careful not just to derive FOCs but to ask whether the equilibrium they describe actually exists and is stable.

4.4 What identifies debt dynamics after a spending shock

The key identifier is the **persistence** of the shock. Because public debt must converge back to its optimal steady-state value, the persistence of the spending process determines how costly it is to defer taxation into the future. This is what makes debt procyclical for transitory shocks and countercyclical for persistent ones.

4.5 What identifies social welfare weights in the quantitative model

The inverse-optimal approach uses the observed steady-state US fiscal system as a revealed-optimal allocation. Given the planner's FOCs, the model backs out the weights q_f consistent with those observed fiscal instruments and allocations.

Why this is credible. It avoids relying on arbitrary utilitarian weights that would generate an unrealistic steady state. Instead, the dynamics are computed around a quantitatively relevant fiscal system.

4.6 Relation to the literature

- Relative to **Aiyagari (1995)**, the paper extends the focus from steady-state public debt to a richer tax system with capital taxes, progressivity, and aggregate shocks.
- Relative to **Chamley–Judd**, it clarifies why zero long-run capital taxation is not generic once incomplete markets and the right utility structure are introduced.
- Relative to recent **HANK Ramsey** work, it contributes existence conditions, debt dynamics, and a practical computational strategy for optimal fiscal policy with many instruments.

5 Tables and figures

The paper has two main tables and three main figures. Their role is mostly quantitative validation rather than reduced-form identification.

5.1 Table 1: Model calibration targets and counterparts

Read:

- The three ex ante types are disciplined by education-related earnings differences in the US.
- The calibrated productivity processes are intentionally heterogeneous but not wildly so: persistence is high for all three types, and the lowest-productivity type faces the greatest idiosyncratic risk.
- This table tells you that the cross-type heterogeneity used in the quantitative model is economically realistic rather than chosen only to force the theory to work.

TABLE 1
MODEL CALIBRATION: TARGETS AND MODEL COUNTERPARTS

	Type 1	Type 2	Type 3
Persistence ρ_y	.986	.98	.98
Variance σ_y	.16	.132	.132
Average productivity	.8	1.0	2.0

TABLE 2
PARAMETER VALUES IN THE BASELINE CALIBRATION

Parameter	Description	Value
	A. Preference and Technology	
β	Discount factor	.992
α	Capital share	.36
δ	Depreciation rate	2.5%
$\bar{a}$	Credit limit	0
χ	Scaling parameter labor supply	.05
φ	Frisch elasticity labor supply	.5
	B. Tax System	
τ^K	Capital tax	36%
κ	Scaling of labor tax	.75
τ	Progressivity of tax	18%
B/Y	Public debt	61%
G/Y	Public consumption	17%

NOTE.—See text for descriptions and targets.

5.2 Table 2: Baseline calibration parameters

Read:

- The steady-state calibration is tightly anchored to US fiscal and macro quantities: $\beta = 0.992$, $\alpha = 0.36$, $\delta = 2.5\%$, $J = 0.5$, capital tax 36%, labor-tax progressivity 18%, debt-output ratio 61%, and public-consumption share 17%.
- This is the quantitative backbone of the inverse-optimal exercise. The model is not trying to match an abstract economy; it is built around a realistic fiscal steady state.
- The high capital tax in the baseline is not an arbitrary theoretical curiosity. It is literally the observed target in the calibration and is then rationalized by the estimated social weights.

5.3 Figure 1: Dynamics of fiscal instruments for high- and low-persistence spending shocks

Read:

- Panel 1 shows that the transitory shock is large on impact but dies quickly, while the persistent shock is much smaller initially because the paper normalizes both to have the same net present value of public spending.
- Panels 2–4 show that the planner responds in both cases by raising the capital tax and increasing labor-tax progressivity, while slightly lowering the average labor-tax level.
- Panel 5 is the key result: public debt rises sharply when persistence is low but falls when persistence is high.
- Panel 6 shows the Lagrange multiplier on the government budget. The shadow value of public resources rises after the shock and remains elevated longer under high persistence.

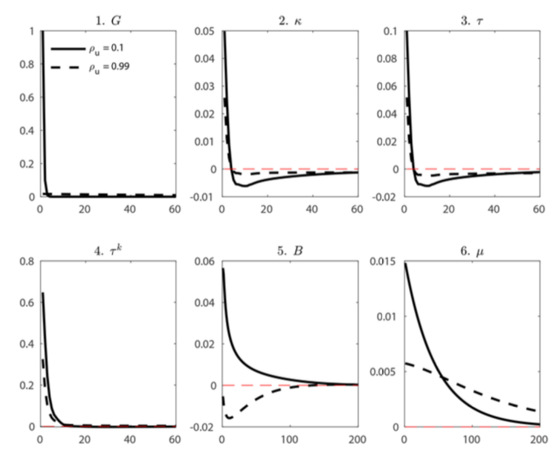

FIG. 1.—Dynamics of selected variables for two shocks with different persistences and same NPV. G, public spending; μ, value of public resources; κ, level of labor tax; τ, progressivity of labor tax; τ^k, capital tax; B, public debt. The solid lines correspond to persistence $\rho_G = 0.1$, and the dashed lines correspond to persistence $\rho_G = 0.99$. G is in percent of GDP, B is in proportional deviations, and other variables are in level deviations.

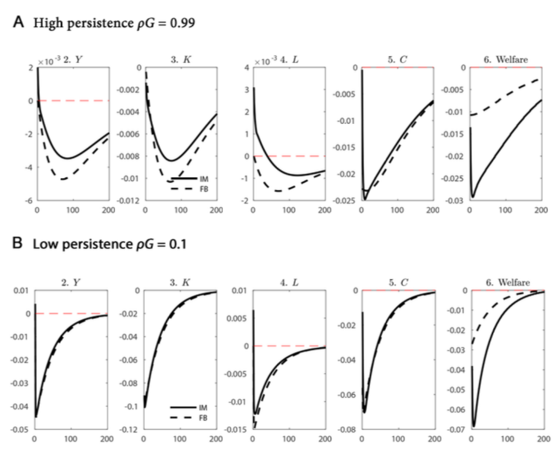

FIG. 2.—Output Y, capital K, labor L, consumption C, and period aggregate welfare (in equivalent consumption) for low and high persistence values, in proportional deviations. The solid line shows the incomplete market model (IM) and the dashed line the first-best allocation (FB). The shock is a public spending shock (panel 1 of app. fig. 5) and differs only in persistence.

5.4 Figure 2: Incomplete markets versus the first best

Read:

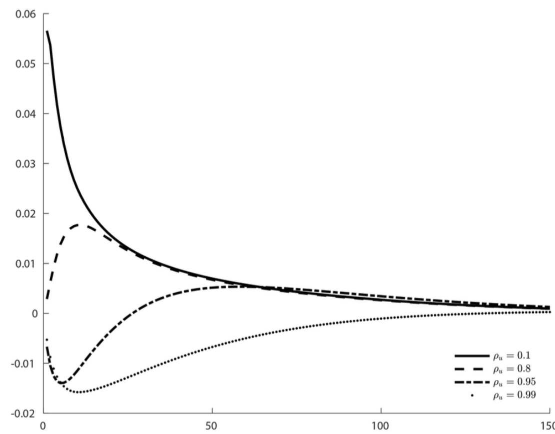

FIG. 3.—Optimal public debt dynamics for different persistence values of public shock (same NPV of public spending), in proportional deviation from steady-state value of public debt.

- The figure compares output Y, capital K, labor L, consumption C, and welfare after the same spending shock in the incomplete-markets economy and a first-best benchmark.
- For both high- and low-persistence shocks, incomplete markets generate more volatile and more persistent responses than the first best.
- Welfare losses are much larger under incomplete markets, especially when the shock is persistent.
- This figure visually explains why debt policy is harder in HANK than in the representative-agent benchmark: self-insurance and distributional concerns amplify the cost of fiscal adjustments.

5.5 Figure 3: Public-debt dynamics for different persistence levels

Read:

- This is the cleanest picture of the paper's main dynamic result. Holding the NPV of public spending fixed, the impact response of debt declines monotonically with persistence.
- For very transitory shocks ($\rho_G = 0.1$), debt jumps up by roughly five to six percent of its steady-state level and then declines.
- For more persistent shocks ($\rho_G = 0.8$ or 0.95), debt initially falls, and the shape changes from inverted-U to J-shaped dynamics.
- The figure makes the paper's financing message intuitive: the more persistent the shock, the more adjustment should come through taxes and the less through debt.

5.6 Robustness and appendix results

Read:

- The core debt-persistence result survives under an alternative affine labor-tax system with lump-sum transfers.

- It also survives under an alternative welfare criterion where weights depend on current productivity rather than ex ante type.
- For TFP shocks, the same qualitative logic as the public-spending shock emerges: debt rises for low persistence and falls for high persistence.
- Discount-factor shocks behave differently because they directly raise desired saving. In that case debt generally falls, though with very persistent discount-factor shocks it can even rise slightly on impact.

6 Short summary

- **Conclusion.** In incomplete-markets economies with occasionally binding credit constraints, positive optimal capital taxes and positive public debt can coexist in steady state.
- **Main theory insight.** The planner uses the capital tax to correct a savings-induced price externality on factor prices, but whether this happens depends strongly on utility curvature and labor-supply structure.
- **Main dynamic insight.** The persistence of public spending shocks determines whether debt should rise or fall on impact: debt smooths transitory shocks but is too costly for persistent ones.
- **Quantitative lesson.** Around a realistic US steady state, optimal fiscal policy after a spending shock raises capital taxes and labor-tax progressivity, while the debt response depends sharply on shock persistence.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes three connected results. First, the steady-state optimal capital tax in a HANK economy can be positive, but not generically: the result requires occasionally binding credit constraints and preference structures such as GHH or DRRA that make the savings externality nonzero. Second, the existence of such an equilibrium requires nontrivial conditions—non-first-best, Straub–Werning, Laffer, and Blanchard–Kahn—so one cannot jump directly from first-order conditions to policy conclusions. Third, in dynamics, public debt is not mechanically a smoothing device: whether it rises or falls after a spending shock depends on persistence.

7.2 Economic intuition

Why can capital taxes be positive here? Because private savings are not just a store of value; they are also a self-insurance device. When households save more, the planner gets a larger capital-tax base, but factor prices also move. Those price changes affect redistribution and the value of self-insurance across heterogeneous households. The planner therefore does not simply maximize the tax base; it trades off tax revenues against the general-equilibrium price effects of savings.

Why can debt fall after a spending shock? Because debt must eventually be stabilized. If the spending shock is persistent, higher debt today means higher taxes tomorrow exactly when capital and output remain depressed. The planner then prefers to front-load adjustment through taxes rather than postpone it through debt.

7.3 Takeaway

This paper is best read as a bridge between classical optimal-tax theory and modern HANK fiscal analysis. Its most important lesson is that incomplete markets do not just change magnitudes; they change the *logic* of fiscal policy. Capital taxation, debt issuance, and labor-tax progressivity become joint tools for managing self-insurance, redistribution, and dynamic tax distortions. For research on fiscal stabilization, debt sustainability, or the incidence of capital taxation in heterogeneous-agent economies, this is a benchmark paper.